VoiceOps 2026: ZentroMart Support Agent

Capstone implementation report | Individual submission | Updated 22 October 2026

Public repository	https://github.com/priyam-chakraborty/voiceops-2026-zentromart
Core stack	Python 3.13.6, Gemini API for generation and embeddings, local retrieval, synthetic course dataset and mock tools, Gnani.ai STT and TTS.
Batch command	`python voice_batch.py --input-dir input_audio --output-dir voice_outputs`

1. Problem and solution

ZentroMart is a fictional Indian online store that receives recurring questions about deliveries, returns, refunds, cancellations, and warranties. This project implements a support assistant intended to answer from company documents, handle selected multi-turn conversations, verify callers before revealing order details, create support tickets when required, and support voice input/output. Gemini is used for language generation and embeddings; Gnani.ai is used for both speech-to-text (STT) and text-to-speech (TTS).

The local RAG index successfully embedded 84 company-document sections. The project reads the supplied synthetic dataset and mock tools. A deterministic exact-match lookup was added for product SKU or product-title questions so that structured product facts can supplement semantic policy retrieval.

2. Implementation and observed tests

Stage	Implementation and observed behavior
Step 1: Prompt-only LLM	Implemented and launched manually. Because this version has no company-document retrieval, policy-specific answers can be incomplete or unsupported. The practice evaluation remains limited by API errors.
Step 2: RAG and product data	Indexed 84 knowledge-base sections. The grocery-return test retrieved RET-5, RET-2 and DWM-1 and stated the change-of-mind restriction plus the 48-hour reporting rule for eligible damaged/expired/wrong/missing items. Unsupported lifetime screen-replacement claims were not confirmed. After an exact-match catalog lookup was added, SKU ELE-0012 was manually answered using products.csv plus WAR-2/WAR-3 evidence, including its ZentroMart Assured status and 12-month warranty.
Step 3: Conversation and tools	Manual tests demonstrated caller verification using the last four phone digits, order lookup, and ticket creation. In a synthetic damaged-TV scenario (INR 31,649), the assistant escalated for human review instead of approving replacement itself, retrieved DWM-3 and ESC-1/ESC-2, and reported ticket creation only after tool success. These selected tests do not establish that all 23 practice conversations pass.
Step 4: Gnani voice	The official gnani-vachana SDK was installed and STT/TTS were tested. The batch runner processed two audio files successfully in one run. A generated grocery-return question completed transcription, RAG retrieval, answer generation, evidence capture and WAV generation; the reply WAV was played and checked as audible. The input question was generated with TTS because the development laptop had no microphone, so this is not a live-microphone or spontaneous-human-speech test.

Local checks on 22 Oct: PASS for `python run.py check` (9 CSVs, 21 KB documents, 1,000 orders, 1,783 order items, 244 support logs, 52 questions, 23 conversations) and for `python -m py_compile run.py voice_batch.py evaluate.py gnani_voice_test.py`.

3. Preliminary evaluation

The included evaluation helper uses transparent string/behavior heuristics, not official course scoring. In an earlier limited run of five practice questions, it reported 2/5 Step 1 heuristic passes (40%) and 3/5 Step 2 passes (60%). For Step 2, the helper reported gold knowledge-base evidence hits for 2/2 cases that specified gold KB section IDs. These results cover only a small sample and should not be treated as a reliable estimate of unseen-test accuracy.

A subsequent five-question run was attempted after the product-catalog improvement. Gemini returned HTTP 503 (temporary unavailability) and HTTP 429 (free-tier request quota exhausted). The helper summary showed 1/5 Step 1 passes and 0/5 Step 2 passes, but three Step 1 calls failed and all five Step 2 calls encountered service/rate errors. Therefore, that rerun is inconclusive and the Step 2 zero is not a valid measure of RAG answer accuracy. No additional API retries were made after quota exhaustion.

Run	Step 1 helper result	Step 2 helper result	Interpretation
Earlier limited run	2/5 passed	3/5 passed; 2/2 gold-evidence hits when specified	Preliminary signal only; sample size is five.
Later rerun	1/5 shown; 3 requests errored	0/5 shown; all 5 requests errored	Inconclusive due to Gemini 429/503 errors; do not interpret as a quality score.

4. Limitations and next improvements

- **Evaluation:** The complete 52-question set and all 23 multi-turn conversations have not been systematically evaluated. API quota availability prevented a reliable comparison after the latest code change.
- **Voice:** The end-to-end batch path was tested on generated audio; live microphone capture and spontaneous human speech were not tested on the development laptop.
- **Reproducibility:** A clean-machine setup/run-through remains useful. The documented batch command is `python voice_batch.py --input-dir input_audio --output-dir voice_outputs`.
- **Next work:** When quota allows, evaluate more practice questions; manually exercise more follow-ups, verification gates and ticket outcomes; then make only evidence-backed fixes.

5. Submission and data handling

The submission archive is intended to include the README, .env.example, requirements, Python entry points, this report, the demo video and the sample voice artifacts. Local archive checks confirmed the expected files, including the exact-match product-catalog fix, and found no .env, virtual environment, cache, dataset copy, or .git history. Only synthetic course data is used, and Gnani.ai is used for both STT and TTS.

Public repository: https://github.com/priyam-chakraborty/voiceops-2026-zentromart. The README documents setup, the exact batch command, environment variables, evaluation helper and data-handling rules.

Scope statement: This report describes only the selected manual tests and limited evaluation runs that were observed. It does not claim that the full practice set passes, that the assistant is production-ready, or that the heuristic percentages are official capstone scores.